

machine, with its array of struts and wires. The first recorded aerial victory is credited to a French aviator. On October 5, 1914, observer Louis Quénault shot down an Aviatik with a Hotchkiss machine gun mounted on a Voisin 8 – a pusher aircraft (with the propeller at the rear). Affording a clear field of fire to the front, pushers were one option for air-combat machines. They proved especially attractive to the British, who introduced the Vickers “Gun bus” in 1915 and the F.E.2 and single-seat D.H.2 pushers the following year. But while pusher machines were by no means ineffective, tractor machines (propeller at the front) were faster and more maneuverable.

What the more skillful and adventurous pilots instinctively yearned for was a gun they could aim simply by pointing their aircraft at the target. Before the war, French and German designers had discovered that it was feasible to create an interrupter gear that would pause a machine gun each time a propeller blade was in its line of fire. Raymond Saulnier, designer of the Morane-Saulnier monoplane, was one of those who experimented with interrupters, but he had not been able to make one work in practice. So it was Dutch designer Anthony Fokker who fitted the first effective interrupter gear to one of his Eindecker monoplanes. The Germans went on to use guns firing through the propeller arc on all their fighters for the rest of the war. Interrupter gears and other forms of synchronizing mechanism tended to reduce the rate of fire of the machine gun, but in later German aircraft, such as the Albatros D.V and Fokker D.VII, the use of twin guns compensated for this drawback.

The Allies’ first effective riposte to the Eindecker’s interrupter gear was to mount a machine gun on the upper wing of a biplane so that it fired over the top of the propeller. Even after the Allies developed their own synchronizing mechanisms to allow firing through the propeller arc, they remained attached to the concept of the wing-mounted gun. Successful solo fighters such as the Nieuport 17 and the S.E.5a were usually equipped with both. In tandem with new armaments, new tactics were also being developed.

Early in the war individual fighters prowled the skies as lone hunters in search of unsuspecting enemy aircraft. By 1916, fighter aircraft were being grouped in squadrons as tactics were developed for fighting in formation. During the titanic battles of Verdun and the Somme, Allied and German airmen fought for air superiority; losses on both sides were heavy in an aerial combat that mirrored the war of attrition on the ground. Numerically inferior,

INVENTING THE FIGHTER PILOT

ON APRIL 1, 1915, FRENCH PILOT Roland Garros positioned his Morane-Saulnier Parasol monoplane behind the tail of a German observation aircraft and fired a burst from his machine gun through the propeller arc. As the German machine plummeted to earth, Garros could claim to have become the first solo fighter pilot. The secret of his success – the ability to fire forward through the propeller – had been achieved by attaching metal plates to deflect any rounds that struck the blades.

Garros had shot down three aircraft by April 18, when engine failure forced him to land behind German lines. His exploits had been highly publicized and the Germans rushed to examine his downed plane. Dutch designer Anthony Fokker was called to Berlin and told to imitate the metal deflectors. Instead, he equipped one of his Eindecker monoplanes (an unarmed reconnaissance aircraft) with an 08/15 Maxim (“Spandau”) machine gun and an interrupter gear copied from a prewar German patent design. The interrupter allowed the pilot to fire through his propeller with less risk to the machine and himself.

German commanders were slow to realize that an important new weapon had been placed in their hands. Eindeckers were introduced in small numbers and were initially spread out in ones and twos, supporting reconnaissance units, which limited their effectiveness. Some German pilots instantly recognized the potential of the new machine. Through the winter of 1915–16, using the simple tactic of swooping down on their enemy from behind in a steep dive, the Eindecker pilots shot down unprecedented numbers of Allied aircraft. The British called it the “Fokker Scourge.” Yet the Eindecker was in fact seriously flawed. It was underpowered, not especially nimble, and had structural



WING-MOUNTED GUN

A Lewis Aerial Gun mounted on the top wing allowed pilots to shoot in the line of flight, over the propeller. The gun was fired by pulling on a cord that led down into the cockpit. The pilot’s problem with this arrangement was changing the ammunition drum in flight, a perilous moment requiring him to take his hands off the controls.

weaknesses. With careful handling it was effective, but it could be a death trap even for an experienced pilot if he put too much stress on the airframe.

The Allies responded with their own solo fighter aircraft. In July 1915, France introduced the small Nieuport 11, affectionately known as the “Bébé” (“Baby”). Originally designed for racing, this light biplane was fast and extremely maneuverable. Its only major weakness was the single-spar lower wing, which allowed the wings to twist in a dive. Although it lacked a synchronized machine gun, it carried a wing-mounted Lewis machine gun, and virtually drove the Eindeckers from the skies. The scourge had finally been scourged.

BOUNCING BULLETS

A Morane monoplane (far left) displays metal plates on its propeller blades. When its machine gun fired, about one in 10 bullets hit the deflectors, bouncing off. This poster shows a German ace in an Eindecker firing through his propeller. The image captures the essence of the solo fighter pilot – the welding of man and aircraft into a single fighting machine.

